

Exercise C1

Q1. If $a, a_1, a_2, a_3, \dots, a_{2n}, b$ are in A.P. and $a, g_1, g_2, g_3, \dots, g_{2n}, b$ are in G.P. and h is the harmonic mean of a and b , then $\frac{a_1 + a_{2n}}{g_1 g_{2n}} + \frac{a_2 + a_{2n-1}}{g_2 g_{2n-1}} + \dots + \frac{a_n + a_{n+1}}{g_n g_{n+1}}$ is equal to

- (A) $\frac{2n}{h}$ (B) $2nh$ (C) nh (D) $\frac{n}{h}$

Ans. A

Sol. $a, a_1, a_2, \dots, a_{2n}, b$ are in AP and $a, g_1, g_2, \dots, g_{2n}, b$ are in GP and $h = \frac{2ab}{a+b}$

$$\therefore \frac{a_1 + a_{2n}}{g_1 g_{2n}} + \frac{a_2 + a_{2n-1}}{g_2 g_{2n-1}} + \dots + \frac{a_n + a_{n+1}}{g_n g_{n+1}} + \dots + \frac{a+b}{ab} + \frac{a+b}{ab} + \dots + \frac{a+b}{ab} = 2n \cdot \left(\frac{a+b}{ab} \right) = \frac{2n}{h}$$

Q2. If $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$, then value of $1 + \frac{3}{2} + \frac{5}{3} + \dots + \frac{2n-1}{n}$ is

- (A) $2n - H_n$ (B) $2n + H_n$ (C) $H_n - 2n$ (D) $H_n + n$

Ans. A

Sol. $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$

$$1 + \frac{3}{2} + \frac{5}{3} + \dots + \frac{2n-1}{n} = (2-1) + \left(2 - \frac{1}{2}\right) + \left(2 - \frac{1}{3}\right) + \dots + \left(2 - \frac{1}{n}\right) = 2n - H_n$$

Q3. If S_1, S_2, S_3 are the sums of first n natural numbers, their squares, their cubes respectively, then $\frac{S_3(1+8S_1)}{S_2^2}$ is equal to

- (A) 1 (B) 3 (C) 9 (D) 10

Ans. C

$$\text{Sol. } \frac{S_3(1+8S_1)}{S_2^2} = \frac{\left[\frac{n(n+1)}{2}\right]^2 \left[1 + \frac{8n(n+1)}{2}\right]}{\left[\frac{n(n+1)(2n+1)}{6}\right]^2} = \frac{[1+4n(n+1)]9}{(2n+1)^2} = 9$$

Q4. Suppose a, b, c are in A.P. and a^2, b^2, c^2 are in G.P. if $a < b < c$ and $a + b + c = \frac{3}{2}$, then the value of a is

- (A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2\sqrt{3}}$ (C) $\frac{1}{2} - \frac{1}{\sqrt{3}}$ (D) $\frac{1}{2} - \frac{1}{\sqrt{2}}$

Ans. D

Sol. $2b = a + c$ and $b^2 = \pm ac$

case - I

$$\text{If } b^2 = ac \quad \text{and} \quad a + c + b = \frac{3}{2} \quad \Rightarrow b = \frac{1}{2}$$

$$a + c = 1 \quad \Rightarrow \quad ac = \frac{1}{4} \quad \Rightarrow \quad (1 - c)c = \frac{1}{4}$$

$$c^2 - c + \frac{1}{4} = 0 \quad \Rightarrow \quad c = \frac{1}{2} \quad \Rightarrow \quad a = \frac{1}{2}$$

$a = b = c$ so not valid

case -II

$$b^2 = -ac \quad \text{and} \quad b = \frac{1}{2}; \quad a + c = 1 \quad \Rightarrow \quad ac = -\frac{1}{4}$$

$$(1 - c)c = -\frac{1}{4} \quad c^2 - c - \frac{1}{4} = 0 \quad \Rightarrow \quad c = \frac{1 \pm \sqrt{1 + 1}}{2}$$

$$= \frac{1 \pm \sqrt{2}}{2}$$

$$c = \frac{1 \pm \sqrt{2}}{2} \quad \Rightarrow \quad a = \frac{1 - \sqrt{2}}{2}$$

Q5. Let the positive numbers a, b, c, d be in A.P. Then abc, abd, acd, bcd are:

(A) not in A.P./G.P./H.P. (B) in A.P. (C) in G.P. (D) in H.P.

Ans. D

Sol. $a, b, c, d \rightarrow$ in A.P. ; $\frac{a}{abcd}, \frac{b}{abcd}, \frac{c}{abcd}, \frac{d}{abcd} \rightarrow$ A.P.

$$\frac{1}{bcd}, \frac{1}{acd}, \frac{1}{abd}, \frac{1}{abc} \rightarrow \text{A.P.} \quad ; \quad \frac{1}{abc}, \frac{1}{abd}, \frac{1}{acd}, \frac{1}{bcd} \rightarrow \text{A.P.}$$

$abc, abd, acd, bcd \rightarrow$ H.P.

Q6. The first term of an infinite geometric progression is x and its sum is 5. Then

(A) $0 \leq x \leq 10$ (B) $0 < x < 10$ (C) $-10 < x < 0$ (D) $x > 10$

Ans. B

Sol. $\frac{x}{1 - r} = 5 \Rightarrow 5 - 5r = x \Rightarrow r = 1 - \frac{x}{5}$

$$\text{As } |r| < 1 \text{ i.e. } \left|1 - \frac{x}{5}\right| < 1; -1 < 1 - \frac{x}{5} < 1$$

$$-5 < 5 - x < 5 = -10 < -x < 0 = 10 > x > 0$$

i.e. $0 < x < 10$

Q7. In the quadratic equation $ax^2 + bx + c = 0$, if $\Delta = b^2 - 4ac$ and $\alpha + \beta, \alpha^2 + \beta^2, \alpha^3 + \beta^3$ are in G.P. where α, β are the roots of $ax^2 + bx + c = 0$, then

(A) $\Delta \neq 0$ (B) $b\Delta = 0$ (C) $c\Delta = 0$ (D) $\Delta = 0$

Ans. C

Sol. $\alpha + \beta = \frac{-b}{a}, \quad \alpha, \beta = \frac{c}{a} \Rightarrow (\alpha + \beta), \alpha^2 + \beta^2, \alpha^3 + \beta^3$ are in G.P.

then $(\alpha^2 + \beta^2) = (\alpha + \beta)(\alpha^3 + \beta^3) \Rightarrow 2\alpha^2\beta^2 = \alpha\beta^3 + \beta\alpha^3$

$\Rightarrow \alpha\beta[\alpha^2 + \beta^2 - 2\alpha\beta] = 0$ so $\alpha\beta(\alpha - \beta)^2 = 0$

$\Rightarrow \frac{c}{a} \cdot \frac{b^2 - 4ac}{a^2} = 0, \quad a \neq 0 \Rightarrow c \cdot \Delta = 0$

Q8. If $(m + 1)^{\text{th}}, (n + 1)^{\text{th}}$ and $(r + 1)^{\text{th}}$ terms of a non-constant A.P. are in G.P. and m, n, r are in H.P., then the ratio of first term of the A.P. to its common difference is :

(A) $-\frac{n}{2}$ (B) $-n$ (C) $-2n$ (D) $+n$

Ans. A

Sol. -

Q9. If α, β be roots of the equation $375x^2 - 25x - 2 = 0$ and $s_n = \alpha^n + \beta^n$, then

$\lim_{n \rightarrow \infty} \left(\sum_{r=1}^n S_r \right) = \dots\dots\dots$

(A) $\frac{1}{12}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) 1

Ans. A

Sol. -

Q10. The sum of the first n terms of the series $1^2 + 2 \cdot 2^2 + 3^2 + 2 \cdot 4^2 + 5^2 + 2 \cdot 6^2 + \dots$ is $\frac{n(n+1)^2}{2}$ when n is even. When n is odd the sum is-

(A) $\frac{3n(n+1)}{2}$ (B) $\frac{n^2(n+1)}{2}$ (C) $\frac{n(n+1)^2}{2}$ (D) $\left[\frac{n(n+1)}{2} \right]^2$

Ans. B

Sol. $1^2 + 2 \cdot 2^2 + 3^2 + 2 \cdot 4^2 + 5^2 + 2 \cdot 6^2 + \dots n \text{ terms} = \frac{n(n+1)^2}{2}$, when n is even

$1^2 + 2 \cdot 2^2 + 3^2 + \dots 2 \cdot n^2 = n \frac{(n+1)^2}{2} \Rightarrow$ when n is odd $n+1$ is even

$1^2 + 2 \cdot 2^2 + 3^2 + \dots n^2 + 2 \cdot (n+1)^2 = (n+1) \frac{(n+1)^2}{2}$

$1^2 + 2 \cdot 2^2 + 3^2 + \dots n^2 = (n+1) \left[\frac{(n+2)^2}{2} - 2(n+1) \right] = \frac{(n+1)n^2}{2}$

Q11. Let a_n denotes the n th term of an A.P. If $a_2 \cdot a_{12} = 1$ & $a_4 \cdot a_{10} = b$, then all the terms of A.P. are distinct and real for the true set of values of 'b' given by:

- (A) $\left(\frac{9}{25}, \infty\right)$ (B) $(0, \infty)$ (C) $(1, \infty)$ (D) (∞, ∞)

Ans. C

Sol. -

Q12. An A.P. whose first terms is unity , and in which the sum of the first half of any even number of terms of that of the second half of the same number of terms is in constant ratio, then the common difference is d is given by

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. B

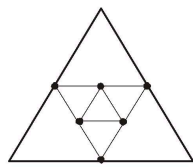
Sol. -

Q13. One side of an equilateral triangle is 24 cm. The mid-points of its sides are joined to form another triangle whose mid - points are in turn joined to form still another triangle. This process continues indefinitely. Then the sum of the perimeters of all the triangles is

- (A) 144 cm (B) 212 cm (C) 288 cm (D) none of these

Ans. A

Sol.



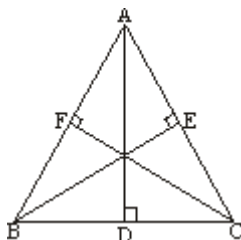
$$= [24 + 12 + 6 + \dots \infty] = 24 \frac{1}{1 - \frac{1}{2}} = 48$$

Q14. If in a ΔABC , the altitudes from the vertices A, B, C on opposite sides are in H.P., then $\sin A$, $\sin B$, $\sin C$ are in :

- (A) G.P. (B) A.P. (C) Arithmetic – Geometric progression (D) H.P.

Ans. B

Sol.



AD, BE, CF are in H.P.

$$\frac{\sin A}{BC} = \frac{\sin B}{AC} = \frac{\sin C}{AB} = 2R$$

In ΔACF

$$\sin A = \frac{CF}{AC} \Rightarrow CF = AC \sin A$$

In $\triangle ABD$

$$\sin B = \frac{AD}{AB} \Rightarrow AD = AB \sin B$$

In $\triangle BCF$

$$\sin C = \frac{BE}{BC} \Rightarrow BE = BC \sin C$$

$$\frac{1}{AD}, \frac{1}{BE}, \frac{1}{CF} \text{ are in A.P.}$$

$$\frac{1}{AB \sin B}, \frac{1}{BC \sin C}, \frac{1}{AC \sin A} \text{ in A.P.}$$

$$\frac{1}{(AB)(2R \cdot AC)}, \frac{1}{(BC)(2R \cdot AB)}, \frac{1}{(AC)(2R \cdot BC)} \text{ are in A.P.}$$

multiply by $2R(AB)(BC)(AC)$

BC, AC, AB are in A.P.

multiply by $2R \Rightarrow 2RBC, 2RAC, 2RAB$ are in A.P.

$\sin A, \sin B, \sin C$ are in A.P.

Q15. A new sequence is obtained from the sequence of positive integers $[1, 2, 3, \dots]$ by deleting all the perfect squares. Then find the 2009th term of the new sequence.

(A) 2052 (B) 2053 (C) 2054 (D) 2055

Ans. C

Sol. -

Q16. Consider $S_n = \frac{8}{5} + \frac{16}{65} + \dots + \frac{8r}{4r^4 + 1}$ Sum of infinite terms of above series will be

(A) 0 (B) $\frac{1}{2}$ (C) 2 (D) None of these

Ans. C

Sol. -

Q17. If the sum of first n terms of an A.P. is cn^2 , then the sum of squares of these n terms is

(A) $\frac{n(4n^2 - 1)c^2}{6}$ (B) $\frac{n(4n^2 + 1)c^2}{3}$ (C) $\frac{n(4n^2 - 1)c^2}{3}$ (D) $\frac{n(4n^2 + 1)c^2}{6}$

Ans. C

Sol. $S_n = cn^2$; $S_{n-1} = c(n-1)^2 = cn^2 + c - 2cn$
 $T_n = 2cn - c$; $T_n^2 = (2cn - c)^2 = 4c^2 n^2 + c^2 - 4c^2 n$

$$\text{Sum} = \sum T_n^2 = \frac{4c^2 \cdot n(n+1)(2n+1)}{6} + nc^2 - 2c^2 n(n+1)$$

$$= \frac{2c^2 n(n+1)(2n+1) + 3nc^2 - 6c^2 n(n+1)}{3} = \frac{nc^2 [4n^2 + 6n + 2 + 3 - 6n - 6]}{3} =$$

$$\frac{nc^2 (4n^2 - 1)}{3}$$

Q18. If x, y, z are in A.P., then $\frac{1}{\sqrt{y} + \sqrt{z}}$, $\frac{1}{\sqrt{z} + \sqrt{x}}$, $\frac{1}{\sqrt{x} + \sqrt{y}}$ will be in :

(A) A.P. (B) G.P. (C) H.P. (D) None

Ans. A

Sol. x, y, z are in A.P.

$$\frac{1}{\sqrt{y} + \sqrt{z}}, \frac{1}{\sqrt{z} + \sqrt{x}}, \frac{1}{\sqrt{x} + \sqrt{y}}$$

$$\frac{\sqrt{y} - \sqrt{z}}{y - z}, \frac{\sqrt{z} - \sqrt{x}}{z - x}, \frac{\sqrt{x} - \sqrt{y}}{x - y}$$

$$x, y, z \Rightarrow x, x + d, x + 2d$$

$$y - z = -d$$

$$z - x = 2d$$

$$x - y = -d$$

$$\frac{\sqrt{y} - \sqrt{z}}{-d}, \frac{\sqrt{z} - \sqrt{x}}{2d}, \frac{\sqrt{x} - \sqrt{y}}{-d}$$

$$\sqrt{z} - \sqrt{y}, \frac{\sqrt{z} - \sqrt{x}}{2}, \sqrt{y} - \sqrt{x}$$

$$(\sqrt{z} - \sqrt{y}) + (\sqrt{y} - \sqrt{x}) = \sqrt{z} - \sqrt{x} = 2 \left(\frac{\sqrt{z} - \sqrt{x}}{2} \right)$$

$$(2b = a + c)$$

$$\Rightarrow \frac{1}{\sqrt{y} + \sqrt{z}}, \frac{1}{\sqrt{z} + \sqrt{x}}, \frac{1}{\sqrt{x} + \sqrt{y}} \text{ are in A.P.}$$

Q19. If $x \in \mathbb{R}$, the numbers $5^{1+x} + 5^{1-x}$, $a/2$, $25^x + 25^{-x}$ form an A.P. then 'a' must lie in the interval:

(A) $[1, 5]$ (B) $[2, 5]$ (C) $[5, 12]$ (D) $[12, \infty)$

Ans. D

Sol. $x \in \mathbb{R}$

$51 + x + 50 \cdot 4^{1-x}$, $a/2$, $5^{2x} + 5^{-2x}$ are in A.P.

$$a = (5^{2x} + 5^{-2x}) + (5^{1+x} + 5^{1-x}) \Rightarrow a = (5^{2x} + 5^{-2x}) + 5(5^x + 5^{-x}) + = (5^x - 5^{-x})^2 + 2 + 5(5^{x/2} - 5^{x/2})^2 + 10$$

$$a = 12 + (5^x - 5^{-x})^2 + 5(5^{x/2} - 5^{x/2})^2 \Rightarrow a \geq 12$$

Q20. If positive numbers a, b, c are in A.P. and a^2, b^2, c^2 are in H.P., then

(A) $a = b = c$ (B) $2b = a + c$ (C) $b^2 = \sqrt{\frac{ac}{8}}$ (D) None of these

Ans. A, B

Sol. -

Q21. Let a and b be two positive real numbers. Suppose A_1, A_2 are two Arithmetic means, G_1, G_2 and H_1, H_2 are Geometric and Harmonic Means between a and b , then

(A) $\frac{G_1 G_2}{H_1 H_2} = \frac{A_1 + A_2}{H_1 + H_2}$ (B) $\frac{G_1 G_2}{H_1 H_2} - \frac{5}{9} = \frac{2}{9} \left(\frac{a}{b} + \frac{b}{a} \right)$ (C) $\frac{H_1 H_2}{A_1 + A_2} = \frac{9ab}{(2a+b)(a+2b)}$
 (D) $\frac{G_1 G_2}{H_1 H_2} = \frac{H_1 + H_2}{A_1 + A_2}$

Ans. A, B, C

Sol. -

Q22. For ΔABC , if $81 + 144a^4 + 16b^4 + 9c^4 = 144abc$, (where notations have their usual meaning), then:

(A) $a > b > c$ (B) $A < B < C$ (C) Area of $\Delta ABC = \frac{3\sqrt{3}}{8}$

(D) Triangle ABC is right angled

Ans. D, C, B

Sol. -

Q23. If $S_r = \sqrt{r + \sqrt{r + \sqrt{r + \sqrt{r + \dots \infty}}}}$, $r > 0$, then which of the following is/are correct.

(A) S_2, S_6, S_{12}, S_{20} are in A.P. (B) S_4, S_9, S_{16} are irrational
 (C) $(2S_3 - 1)^2, (2S_4 - 1)^2, (2S_5 - 1)^2$ are in A.P. (D) S_2, S_{12}, S_{56} are in G.P.

Ans. D, C, B, A

Sol. -

Q24. If a, b, c are in H.P., then:

- (A) $\frac{a}{b+c-a}, \frac{b}{c+a-b}, \frac{c}{a+b-c}$ are in H.P. (B) $\frac{2}{b} = \frac{1}{b-a} + \frac{1}{b-c}$
 (C) $a - \frac{b}{2}, \frac{b}{2}, c - \frac{b}{2}$ are in G.P. (D) $\frac{a}{b+c}, \frac{b}{c-a}, \frac{c}{a+b}$ are in H.P.

Ans. D, C, B, A

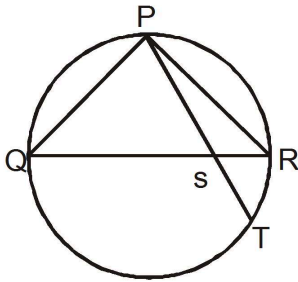
Sol. -

Q25. A straight line through the vertex P of a triangle PQR intersects the side QR at the point S and the circumcircle of the triangle PQR at the point T. If S is not the centre of the circumcircle, then

- (A) $\frac{1}{PS} + \frac{1}{ST} < \frac{2}{\sqrt{QS \times SR}}$ (B) $\frac{1}{PS} + \frac{1}{ST} > \frac{2}{\sqrt{QS \times SR}}$ (C) $\frac{1}{PS} + \frac{1}{ST} < \frac{4}{QR}$
 (D) $\frac{1}{PS} + \frac{1}{ST} > \frac{4}{QR}$

Ans. B, D

Sol.



$$PS \cdot ST = QS \cdot SR$$

Now HM < GM

$$\Rightarrow \frac{1}{ST} + \frac{1}{ST} > \frac{2}{\sqrt{QS \cdot SR}}$$

Now QR = QS + SR

Applying AM > GM

$$\frac{QS + SR}{2} < \sqrt{QS \cdot SR}$$

$$QR > \sqrt{QS \cdot SR} \quad \frac{4}{QR} < \frac{2}{\sqrt{PS \cdot ST}} \Rightarrow \frac{1}{PS} + \frac{1}{ST} > \frac{2}{\sqrt{PS \cdot ST}} > \frac{4}{QR}$$

$\therefore$ D is correct and C is wrong

$\Rightarrow$ B and D are correct